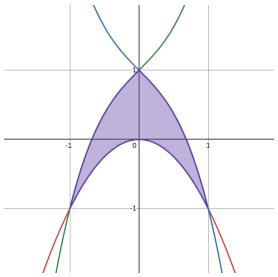
Math 152 Problem Set 4

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Problem 1. Compute the area of the region depicted below, bounded by the following curves.

$$y = -x^2$$
$$y = 1 + x + x^3$$
$$y = 1 - x - x^3$$

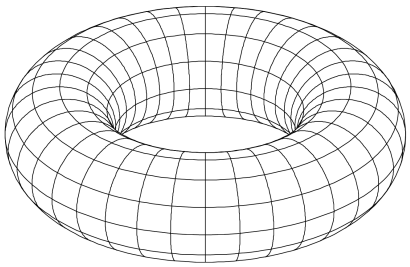


Problem 2. Evaluate the following definite integral.

$$\int_{-9\pi/2}^{-7\pi/2} \sin(x)^4(2\sin(x) + \cos(x))\mathrm{d}x$$

Problem 3. Suppose a bowl in the shape of a half-sphere contains water coming up to half of its height. What fraction of its *volume* is filled?

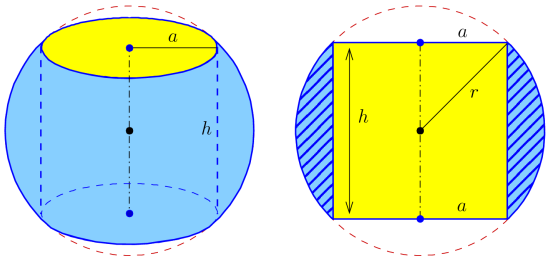
Problem 4. A *solid torus* is a solid obtained by revolving a circular disk about an axis, depicted below.



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Suppose that $0 < r < R$ are two radii. Find the volume of the torus obtained by revolving a disk of radius r centered at the point $(0, R)$ about the y-axis.

Problem 5. Consider a ring obtained by drilling a cylinder out of a sphere of radius r, so that the height of the resulting solid is $h < 2r$.



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(a) Use the Pythagorean Theorem to find the radius a of the center hole in terms of r and h.

(b) Compute the volume of the ring. If you did this right, it should depend only on h and not on r. Isn't that strange?

Problem 6 (Optional)**.** Suppose that f is an odd function. Show that the accumulation function

$$F(x) = \int_0^x f(t)\mathrm{d}t$$

is even.

Problem 7 (Optional)**.** Consider a reservoir, the water evaporating away in the hot sun. Water volume is lost at a rate proportional to the exposed surface area at any given time. Argue that whatever the shape of the reservoir, the height of the surface drops as a constant rate.